Course notes

Matrix sequences

Property · Sequence $U_{n+1} = AU_n + B$. Let A be a square matrix of order p, B a column matrix and (U_n) a sequence of column matrices such that $U_{n+1} = AU_n + B$ for every n. If $I_p - A$ is invertible, the matrix $C = (I_p - A)^{-1}B$ satisfies $C = AC + B$, and for every natural number n:

$$U_n = A^n (U_0 - C) + C.$$

Proof. $(I_p - A)C = B$ can be written $C = AC + B$. Subtracting: $U_{n+1} - C = A(U_n - C)$. The sequence $V_n = U_n - C$ satisfies $V_{n+1} = AV_n$, so $V_n = A^n V_0$ (induction), and $U_n = A^n(U_0 - C) + C$. □

Key point · Behaviour at infinity. If all the entries of A^n tend to 0, then U_n converges to C (entry by entry). This behaviour can be observed with a calculator; it is taken for granted when the statement says so.

Markov chains

Definition · Markov chain with two or three states. A system is observed at times $0, 1, 2, \ldots$; at each time, it is in one of the states $1, \ldots, k$ ($k = 2$ or 3). Let X_n be the state at time n. We say that (X_n) is a (homogeneous) *Markov chain* if, given the present state X_n, the distribution of X_{n+1} does not depend on the earlier states $X_0, \ldots, X_{n-1}$ (Markov property), and if the probability of moving from state i to state j does not depend on n (homogeneity). We then write, for all i, j and n such that $P(X_n = i) \neq 0$,

$$P_{(X_n=i)}(X_{n+1} = j) = t_{ij}.$$

The numbers t_{ij} are the *transition probabilities*.

Definition · Probabilistic graph, transition matrix. The chain is represented by a *probabilistic graph*: a weighted directed graph whose vertices are the states, with an arc from i to j, of weight t_{ij}, when $t_{ij} > 0$. The *transition matrix* is the square matrix $T = (t_{ij})$. Its entries are nonnegative and the sum of each row is 1.

Definition · Distribution. The *distribution* at time n is the row matrix $\pi_n = \begin{pmatrix} P(X_n = 1) & \cdots & P(X_n = k) \end{pmatrix}$; π_0 is the initial distribution.

Property · Evolution of the distribution. For every natural number n: $\pi_{n+1} = \pi_n T$ and $\pi_n = \pi_0 T^n$.

Proof. By the law of total probability, with the partition $(X_n = 1), \ldots, (X_n = k)$ (events of probability zero are left out, as they contribute 0):

$$P(X_{n+1} = j) = \sum_{i=1}^{k} P(X_n = i)\, t_{ij},$$

which is entry j of $\pi_n T$. The second formula follows by induction. □

Property · Transition in n steps. If $P(X_0 = i) = 1$, then for every state j, $P(X_n = j)$ is the entry in row i and column j of T^n. In other words, the probability of moving from state i to state j in n transitions is $(T^n)_{ij}$.

Proof. The initial distribution is the row π_0 with a 1 in position i and 0s elsewhere. Then $\pi_n = \pi_0 T^n$ is row i of T^n. □

Definition · Invariant distribution. A distribution π (a row of nonnegative entries with sum 1) is *invariant* if $\pi T = \pi$. If $\pi_0 = \pi$, then $\pi_n = \pi$ for every n.

Property · Two-state case. Let $T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $0 \leq a \leq 1$, $0 \leq b \leq 1$ and $0 < a + b < 2$. The chain has a unique invariant distribution $\pi = \begin{pmatrix} \frac{b}{a+b} & \frac{a}{a+b} \end{pmatrix}$, and whatever the initial distribution, π_n converges to π.

Proof. Write $\pi_n = \begin{pmatrix} x_n & 1-x_n \end{pmatrix}$. Then $x_{n+1} = (1-a)x_n + b(1-x_n) = (1-a-b)x_n + b$. This arithmetico-geometric sequence has fixed point $\ell = \frac{b}{a+b}$, and $x_n - \ell = (1-a-b)^n(x_0 - \ell)$. Since $-1 < 1-a-b < 1$, $(1-a-b)^n \to 0$ and $x_n \to \ell$. An invariant distribution satisfies $x = (1-a-b)x + b$, so $x = \ell$: it is unique. □

Method · Three states: invariant distribution. Solve the system formed by $\pi T = \pi$ and the condition $a + b + c = 1$; one of the three equations of $\pi T = \pi$ follows from the other two. Possible convergence of π_n is observed with a calculator.

Watch out. The distribution is a row and we multiply on the right by T: $\pi_{n+1} = \pi_n T$, not $T\pi_n$. It is the rows of T, not its columns, that add up to 1.